Mahmoud Alyosify

AI Engineer | LLMs, RAG & AI-Powered Software

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SUMMARY

AI Engineer with an **MSc in Artificial Intelligence from Queen's University, Canada (GPA 4.30 / 4.30)**. Led a master's thesis project mentored by **Deloitte's AI team** that makes LLMs answer with 47.6% fewer tokens, published as two first-author papers on arXiv. Experienced in building RAG assistants, AI agents, fine-tuned LLMs and machine learning models, from the data to a working, tested API. Currently an AI Engineer Intern at Injaz Tech. Also has a full-stack background in .NET and Angular, and six years of teaching AI and computer science.

SKILLS

AI & LLMs: RAG, LangChain, LangGraph, AI agents, Hugging Face, fine-tuning (LoRA, QLoRA), prompts
Machine learning: PyTorch, TensorFlow, scikit-learn, XGBoost, computer vision, NLP
Data & cloud: Python, SQL, Pandas, FAISS, AWS, Docker
Software: C#, ASP.NET Core, Entity Framework, SQL Server, Angular, TypeScript, FastAPI, Git

EXPERIENCE

AI Engineer Intern — Oct 2026 – Present
Injaz Tech (Injaz Group, formerly Fingerprint Consultancy) — Maadi, Cairo

- Building a RAG assistant over company documents with **LangChain, LangGraph, FAISS** and **FastAPI**.
- Raised top-1 retrieval accuracy from **75% to 94%** and cut wrong refusals from 5 in 20 to 1 in 20.
- Added **136 automated tests** and an evaluation check for every new version.

Project Lead — MinimalLM (Master's Thesis) — Jun 2026 – Oct 2026
Queen's University · industry-mentored by Deloitte's AI team

- Led the team that built a training-free method that makes open LLMs (Mistral, Llama 3.1, Qwen 2.5) give shorter answers that are still complete, mentored throughout by **Deloitte's AI team**.
- On 330 questions: **47.6% fewer tokens**, **19–68% faster** answers and no answer cut off. Published as two first-author papers on arXiv.

Part-Time Machine Learning Instructor — Jul 2025 – Present
National Telecommunication Institute (NTI) — Egypt

- Teach **90-hour machine learning bootcamps** and AI for Business modules.

AI in Cybersecurity Intern — Jun 2025 – Jul 2025
e& Egypt (on-the-job training via NTI) — Cairo

- Used AI tools for threat intelligence, digital forensics and security analysis.

Systems Engineer (Military Service) — Feb 2024 – Mar 2025
National Company for Roads Building & Development — Cairo

- Supported the backend of Egypt's national toll network, used by **50%+ of the country's vehicles**.
- Co-developed a data-management and reporting app in **C#, .NET, Entity Framework, SQL Server**.

Online Instructor — 2020 – Present
Udemy & YouTube (Arabic courses) — Remote

- Arabic courses on algorithms, discrete math, probability and statistics: **10,000+ learners, 700K+ views**.

EDUCATION

MSc in Artificial Intelligence — Sep 2025 – Oct 2026
Queen's University — Kingston, Canada

- GPA **4.30 / 4.30** · **Digilians Presidential Scholarship** (competitive national AI fellowship)
- Thesis: MinimalLM — shorter LLM answers that stay complete (supervisor: Dr. Rahatara Ferdousi)

BSc in Computer & Information Science (Bioinformatics) — Sep 2019 – Jul 2023
Assiut University — Assiut, Egypt

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+

PUBLICATIONS

Say Less, Mean Better: Training-Free Verbosity Control for Language Models
M. Alyosify, M. Imbabi, E. Elkhamisi, R. Ferdousi · arXiv:2611.04107 (cs.CL) · November 2026

Beyond Truncation? Entailment-Licensed Elision and the Limits of Semantic Preservation in LLM Decoding
M. Alyosify, M. Imbabi, E. Elkhamisi · arXiv:2611.08975 (cs.CL) · November 2026

PROJECTS

HORUS Sentinel — multi-agent threat-intelligence platform 2026
Python, LangGraph, FastAPI, Neo4j, Ollama, Docker

- A team of AI agents collects open-source information into a **Neo4j knowledge graph**; a self-hosted, fine-tuned Llama-3-8B writes evidence-backed reports for analysts.

Horus-OSINT — fine-tuned LLM on AWS 2026
Python, Llama-3-8B, QLoRA, PySpark, AWS (EC2, S3, EMR), Terraform

- Fine-tuned **Meta-Llama-3-8B** with QLoRA on **159,826** GTD and GDELT records and deployed it on **AWS** as a threat-intelligence chatbot.

SCRAP — satellite collision risk prediction 2026
Python, XGBoost, LightGBM, Optuna, SHAP

- Predicts satellite collision risk **48+ hours** ahead: **97.31% recall** on high-risk events. Interactive 3D demo online.

Automated Multimodal Agent — PDF to narrated slides 2026
Python, LangGraph, Hugging Face, FAISS, Streamlit, text-to-speech

- Turns a PDF into a narrated PowerPoint deck (8-step LangGraph pipeline). Presented at the AI in Teaching and Learning Forum 2026.

SimCLR-Vision-SSL — self-supervised image model 2026
PyTorch, SimCLR, SupCon, ONNX, FAISS

- Trained an image model without labels: 42 experiments, **84.30% top-1 accuracy**, image search with ONNX and FAISS.

Vitalism Solution — graduation project 2023
Python, computer vision, signal processing (rPPG)

- Reads heart rate and other vital signs from ordinary video (rPPG). Grade A+; **Microsoft Imagine Cup 2023 semi-finalist**.

Full-Stack Web Applications — .NET & Angular
C#, ASP.NET Core, Entity Framework, SQL Server, Angular, TypeScript

- Inventory system (ASP.NET Core, Entity Framework, SQL Server): stock tracking, barcodes, low-stock alerts.
- E-commerce web app (Angular, JWT, Stripe): product catalogue, cart, secure login and online payments.

Also: AudioShield (deepfake audio detection) · RL-Cloud-Autoscaler (PPO, DQN) · Human vs. AI-agent pull requests (research study).

CERTIFICATIONS & TRAINING

- AWS Certified Cloud Practitioner (Amazon Web Services)
- IBM AI Developer Professional Certificate (IBM, 2026)
- Deep Learning Specialization and Machine Learning Specialization (DeepLearning.AI, 2026)
- Fortinet Certified Associate in Cybersecurity (2025) · AI in Cybersecurity (NTI, 420 hours, 2025)
- Full-Stack Development Diploma, .NET & Angular (Route, 2023–2024) · ITI .NET and SQL Server courses
- AWS Academy: Generative AI, Data Engineering, ML for NLP · ML Internship, ITIDA (200 hours, 2023)

AWARDS & ACTIVITIES

- 1st Place, Smart Cities Hackathon 2022: a system that monitors ICU patients' vital signs (77 university teams)
- Semi-Finalist, Microsoft Imagine Cup 2023 · Finalist, ISEIC 2023 (800+ teams)
- Ideal Student Award, Faculty of Computers & Information, Assiut University (2022–2023)
- Secretary, Scientific & Technological Committee, Assiut University Student Union · ITI Student Ambassador

LANGUAGES

Arabic (native) · English (professional working proficiency)